

Introduction

Looking at a star's position will not give you the full picture of its life. To fully grasp the history of a star, we can look at its **chemistry** and **kinematics**. From high-resolution spectra of stars, we can get both of these sets of information and assess the formation site of stars in the Milky Way.

There are different scenarios for where stars in the Milky Way were born. Stars can be born *in situ* in the disk or halo. Some stars can be born in the disk and then pulled into the halo. They can also be accreted from dwarf satellite galaxies. To distinguish between these scenarios, we assess the chemical abundances of red giants at the interface of the Milky Way's disk and halo components.

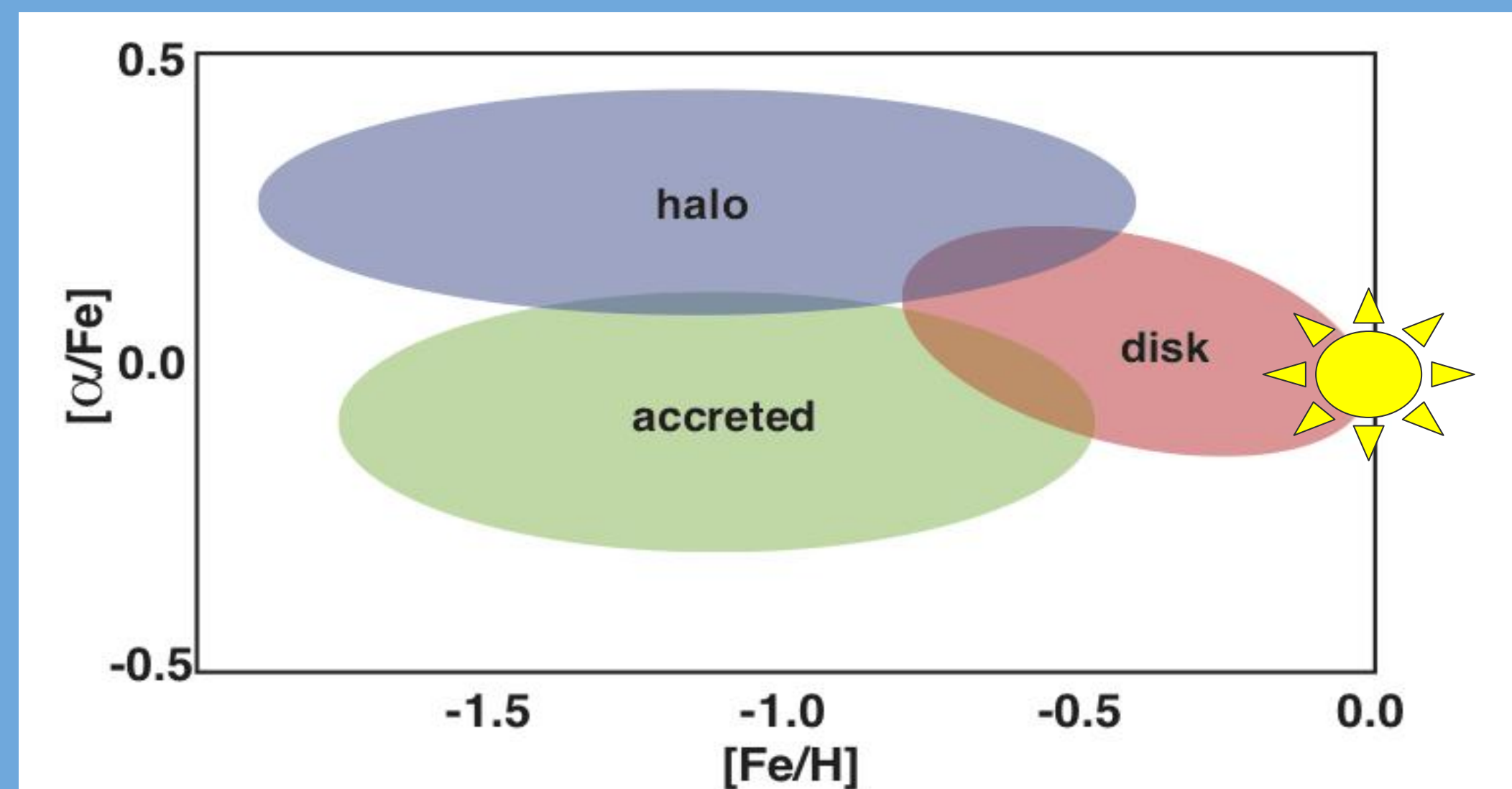


Figure 1: A schematic of the difference in metallicity [and alpha/Fe] of stars born in different situations.

Figure 1 shows a chemical comparison of stars that are born in different parts of a galaxy. One important characteristic we look at is metallicity, which is a measure of how metal-enriched a star is compared to our sun. We also have alpha elements, which we use in addition to metallicity to chemically tag stars.

Methodology

We used a software called Spectroscopy Made Harder (SMHr) which is used to analyze echelle spectra. To do this we measure the strengths of absorption lines from red giant spectra. From there, we iteratively fit stellar atmospheric parameters to existing models. In Figure 2 is an image of a spectra from the red giant star Arcturus, which is one of the stars we looked at in this study. The highlighted bars are examples of two different absorption lines from two different elements. Every element has a unique set of spectral lines at different wavelengths. Thus, we can distinguish various atoms present in a star's atmosphere.

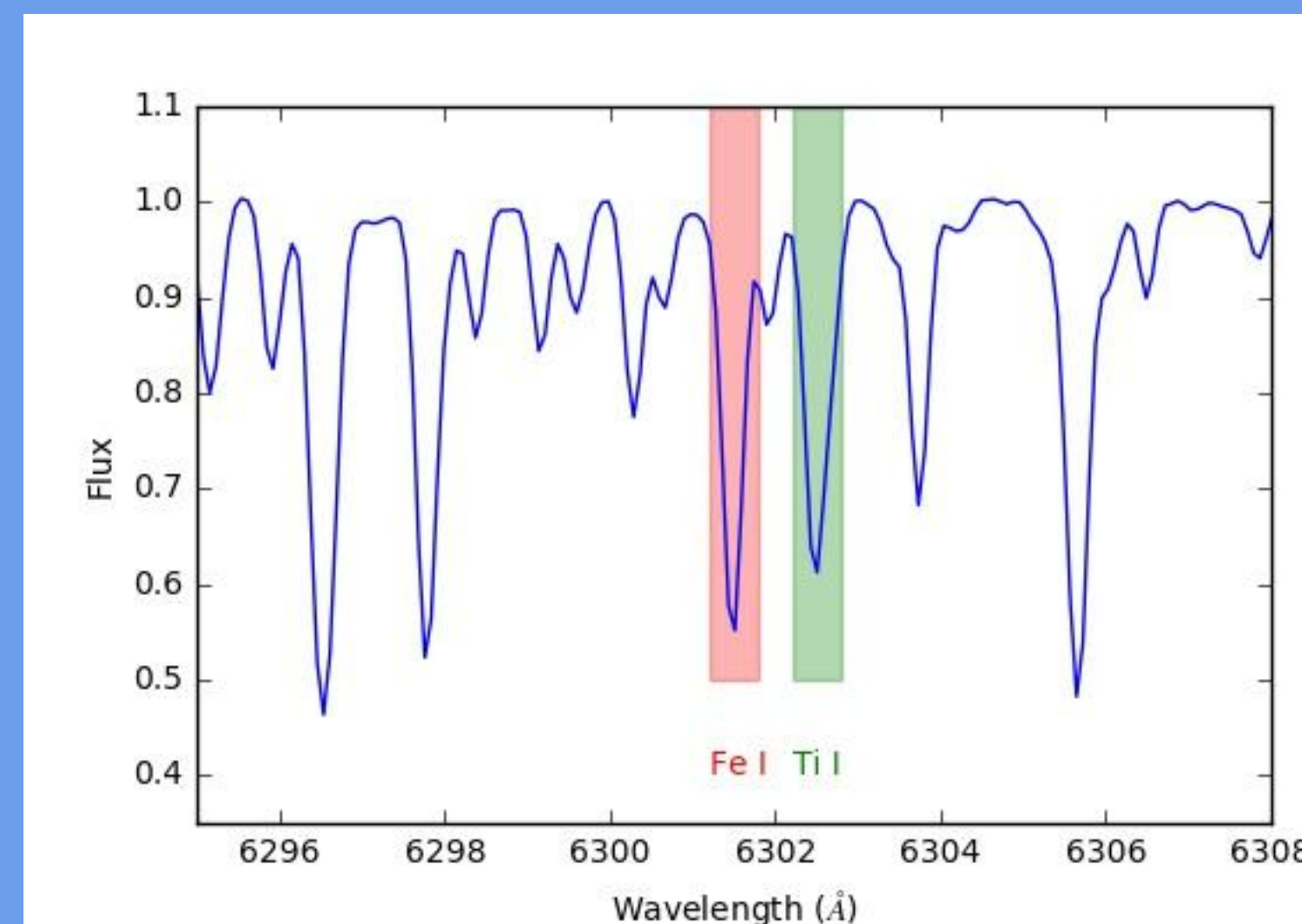


Figure 2: The spectra of Arcturus. The red bar is an absorption line of iron and the green is titanium.

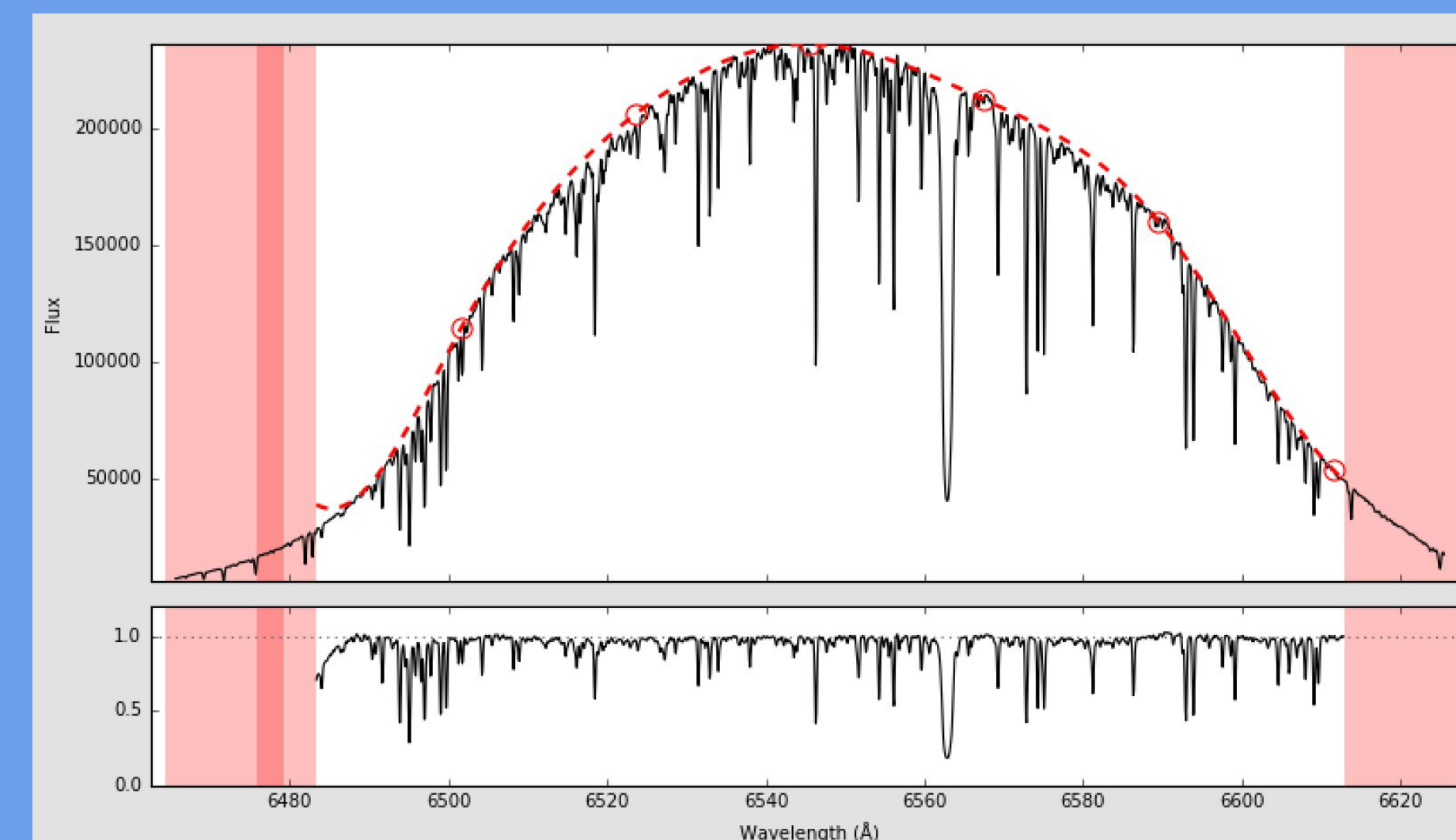


Figure 3: This is an image from SMHr, showing how we process our data and derive stellar atmospheric parameters.

Results

	SMHR	Literature (Carlin, 2018)
Temperature (K)	4300	4350 ± 75
log Gravity	1.6	1.6 ± 0.4
Metallicity	-0.65	-0.58 ± 0.12

Table 1: These are the values that we derived using SMHr compared to values from a previous study done.

These are the results we obtained from SMHr for the red giant standard Arcturus compared to values from previous literature. Our results fall within the error for the literature results, so we have a proof of concept of our methodology.

Using the red giant standard star Arcturus, we verified that our results are consistent with the current literature. After using SMHr on Arcturus, we were able to derive its stellar atmospheric parameters, which are shown in Table 1.

Future Work

Our next steps will be to apply this methodology to a larger set of red giants at the Milky Way's disk-halo interface to distinguish between formation scenarios for these stars.

References

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